

AMAN KUMAR SINGH

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Summary

Full-stack developer specializing in Java, Node.js, React.js, and MongoDB, with experience building secure and scalable web applications. Proficient in REST API development, authentication systems, and backend engineering, with expanding knowledge of Spring Boot. Strong interest in clean architecture, efficient code, and open-source collaboration.

Education

Master of Computer Application

Sharda University, Uttar Pradesh

PG: 2026

CGPA: 9.27 / 10

Bachelor of Computer Application

JIS College of Engineering, West Bengal

UG: 2024

CGPA: 8.14 / 10

Projects

Air Quality Analysis & Forecasting Dashboard | Python, Flask, LSTM

Feb 2026

- Developed a web-based air quality monitoring system integrating real-time environmental data using APIs to track AQI and pollutant levels (PM2.5, PM10, NO₂, CO, O₃).
- Performed data cleaning and exploratory data analysis (EDA) using **Pandas** and **NumPy** to clean, transform, and analyze air quality datasets.
- Built a **LSTM-based deep learning model** to forecast PM2.5 levels for the next 24 hours, enabling predictive analysis of pollution trends.

EchoSafe Journal | Java, Spring Boot - Backend

Mar 2025

- Architected and implemented a **Spring Boot backend with MongoDB**, applying **RESTful** microservices principles to ensure scalability and high availability.
- Developed and enforced JSON Web Token (JWT) & OAuth2 authentication and role-based access control using **Spring Security**.
- Designed, documented, and validated **RESTful APIs** with **JUnit** testing to guarantee scalability, reliability, and maintainability.

ContactVault | React.js, Node.js, Express.js, MongoDB

Mar 2025

- Developed a full-stack contact management application using **React.js**, **Node.js**, **Express.js**, and **MongoDB** with responsive UI components and seamless user experience.
- Built secure RESTful APIs with optimized CRUD operations, implementing **JWT authentication** and **Role-Based Access Control (RBAC)** for secure user access.
- Integrated frontend interfaces with backend APIs, enabling dynamic data rendering, authentication workflows, and efficient client-server communication.
- Improved application scalability and performance through MongoDB indexing, aggregation pipelines, reusable middleware architecture, and optimized API response handling. “

Technical Skills

Languages: Java, C, C++, JavaScript (ES6+), TypeScript, Python, SQL

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design

Frameworks Libraries: Spring Boot, Node.js, Express.js, Redux Toolkit, Zustand (Basics), shadcn/ui

Backend APIs: REST API Development, API Integration, JWT Authentication, Role-Based Access Control (RBAC)

Databases: MySQL, MongoDB

Tools Platforms: AWS, Git, GitHub, Vercel, Postman, IntelliJ IDEA, VS Code, Power BI, Excel

Development Practices: Component-Based Architecture, Responsive Design, Cross-Browser Compatibility, Performance Optimization

Certifications

- **Data Visualization using Power BI** - Sharda Informatics — May–June 2025 [↗](#)
- **Prompt Engineering** - Sharda Informatics — Jan–April 2025 [↗](#)
- **AI: Transformative Learning with TechSaksham** - TechSaksham — 2025 [↗](#)
- **Python Pro Bootcamp** - Udemy — 2024 [↗](#)
- **Mastering in C** - Udemy — 2022 [↗](#)